

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Electrical Engineering

2. Subject Name: Analog Electronic Circuits Laboratory, L-T-P: 0-0-3, Credits: 2

3. Pre-requisites: No

4. Syllabus and reference books:

Syllabus:

Experiment 1: D.C. characterization of BJT and design of simple amplifier

Experiment 2: Design of common emitter (CE) amplifier with stable operating point

Experiment 3: Design of common collector (CC) amplifier and cascaded CE-CC amplifier

Experiment 4: D.C. characterization of MOSFET and design of simple amplifier

Experiment 5: Design of common source (CS) amplifier with stable operating point

Experiment 6: Design of common drain (CD) amplifier and cascaded CS-CD amplifier

Experiment 7: Characterization of basic and cascode current mirror circuits (with BJT and MOSFET)

Experiment 8: Design of BJT and MOSFET differential amplifier with resistive load and active load

Experiment 9: Design of R-C and L-C oscillators (phase shift/Colpitt/Hartley)

Experiment 10: Design of a second order active filter (low pass/high pass)

Experiment 11: Design of tuned amplifier

Reference Books:

1) "Fundamentals of Microelectronics", B. Razavi, 2nd Edition.

2) "Microelectronic Circuits", A. S. Sedra and K. C. Smith, 7th Edition

3) "Analysis and Design of Analog Integrated Circuits", Gray, Hurst, Lewis, and Meyer, 5th Edition

4) "Foundation of Analog and Digital Electronic Circuits", Anant Agarwal and Jeffrey Lang

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	D.C. characterization of BJT and design of simple amplifier	3 hours
2.	Design of common emitter (CE) amplifier with stable operating point	3 hours
3.	Design of common collector (CC) amplifier and cascaded CE-CC amplifier	3 hours
4.	D.C. characterization of MOSFET and design of simple amplifier	3 hours
5.	Design of common source (CS) amplifier with stable operating point	3 hours
6.	Design of common drain (CD) amplifier and cascaded CS-CD amplifier	3 hours
7.	Characterization of basic and cascode current mirror circuits (with BJT and MOSFET)	3 hours
8.	Design of BJT and MOSFET differential amplifier with resistive load and active load	6 hours
9.	Design of R-C and L-C oscillators (phase shift/Colpitt/Hartley)	3 hours
10.	Design of a second order active filter (low pass/high pass)	3 hours
11.	Design of tuned amplifier	3 hours
Total number of hours		36 hours